# Variance-Adaptive Approximated Model Predictive Control

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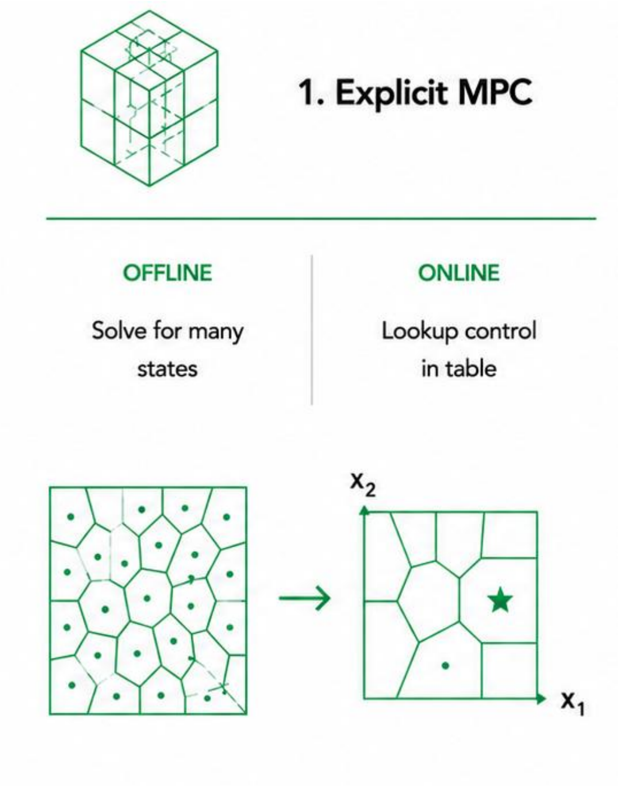
European Control Conference (ECC) 2026, Reykjavík, Iceland

# How to Remove Online Optimization?



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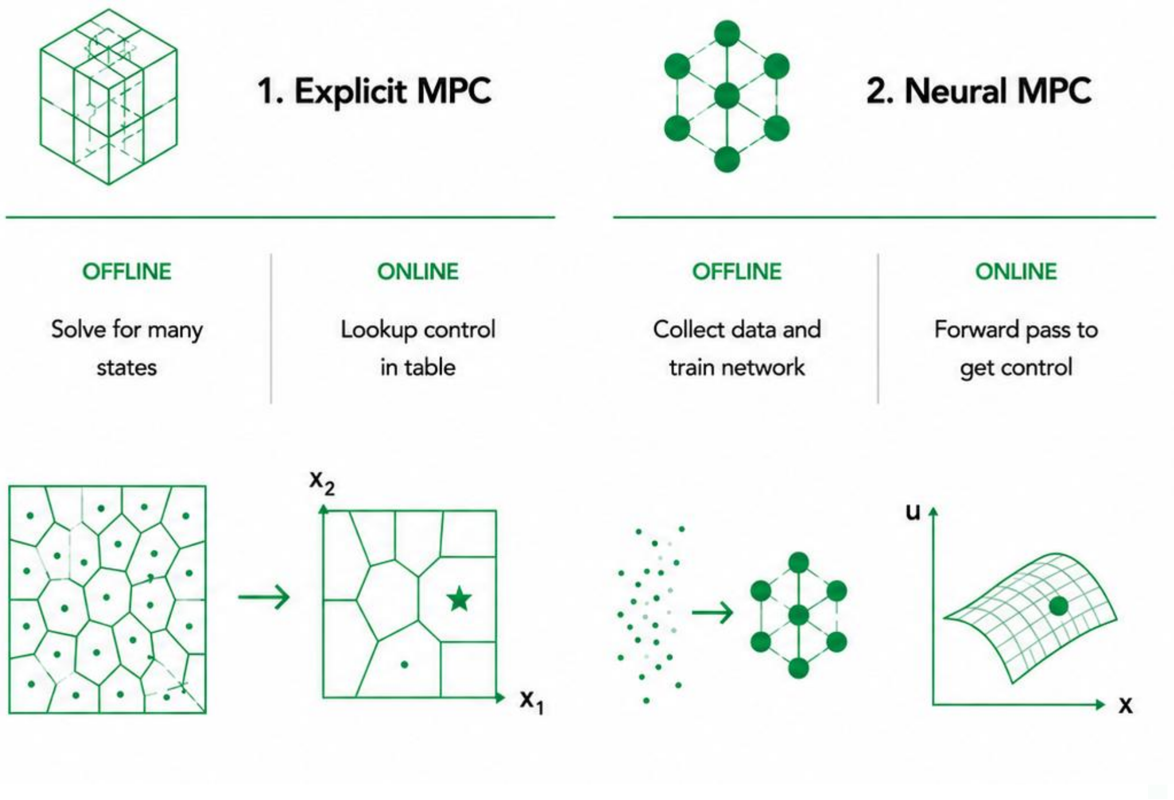

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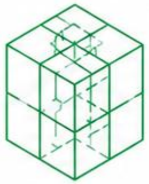


## 1. Explicit MPC

**OFFLINE**

Solve for many states

**ONLINE**

Lookup control in table

## 2. Neural MPC

**OFFLINE**

Collect data and train network

**ONLINE**

Forward pass to get control

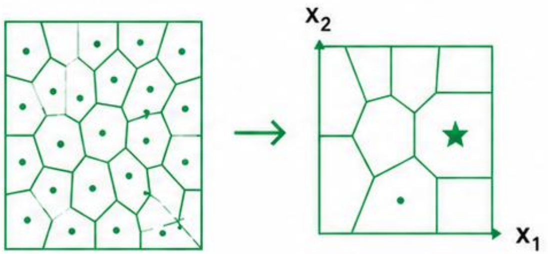
## 3. Sampling-based MPC

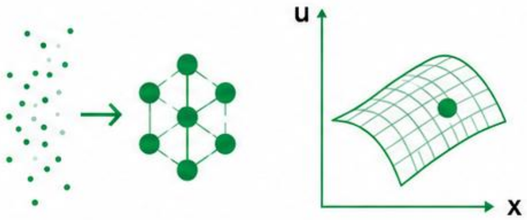
**OFFLINE**

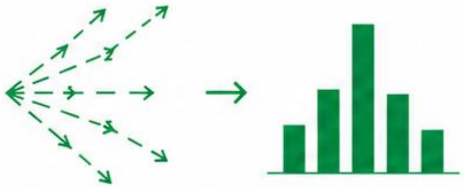
No offline phase

**ONLINE**

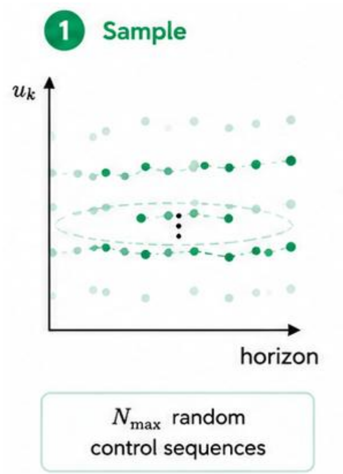
Sample, simulate, and select the best control sequence



# Random-Shooting MPC



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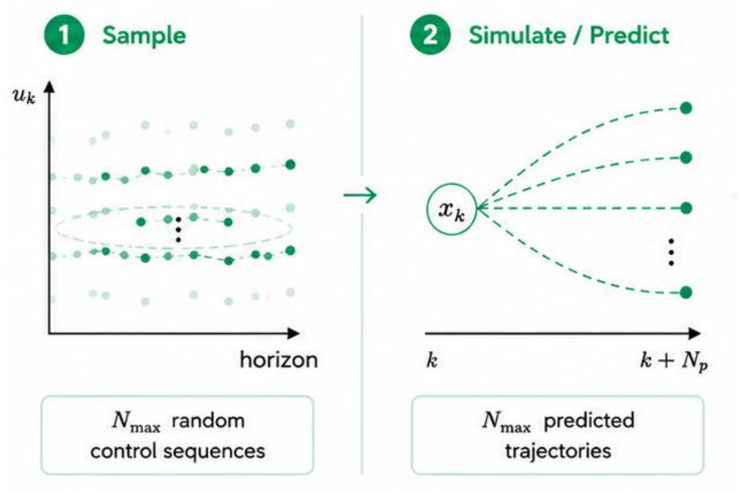

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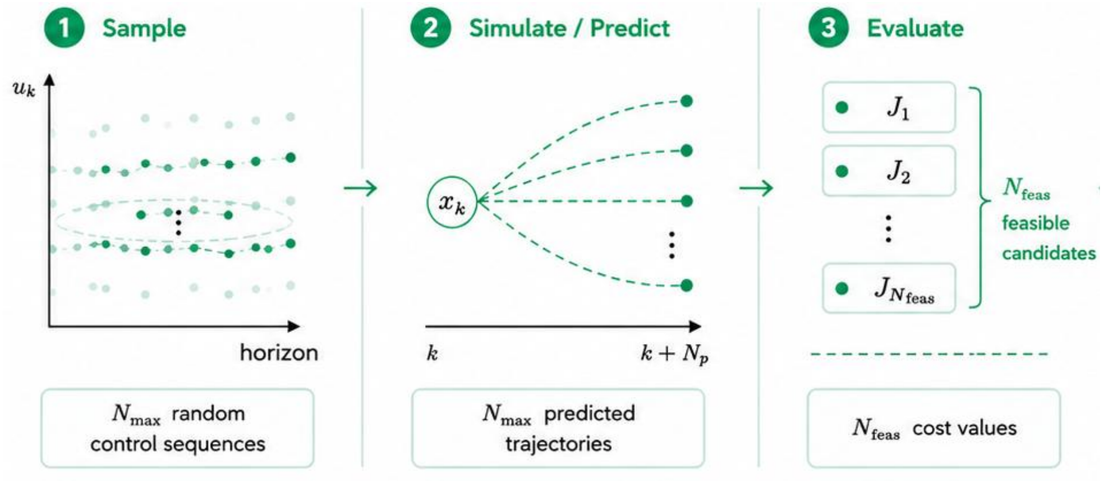

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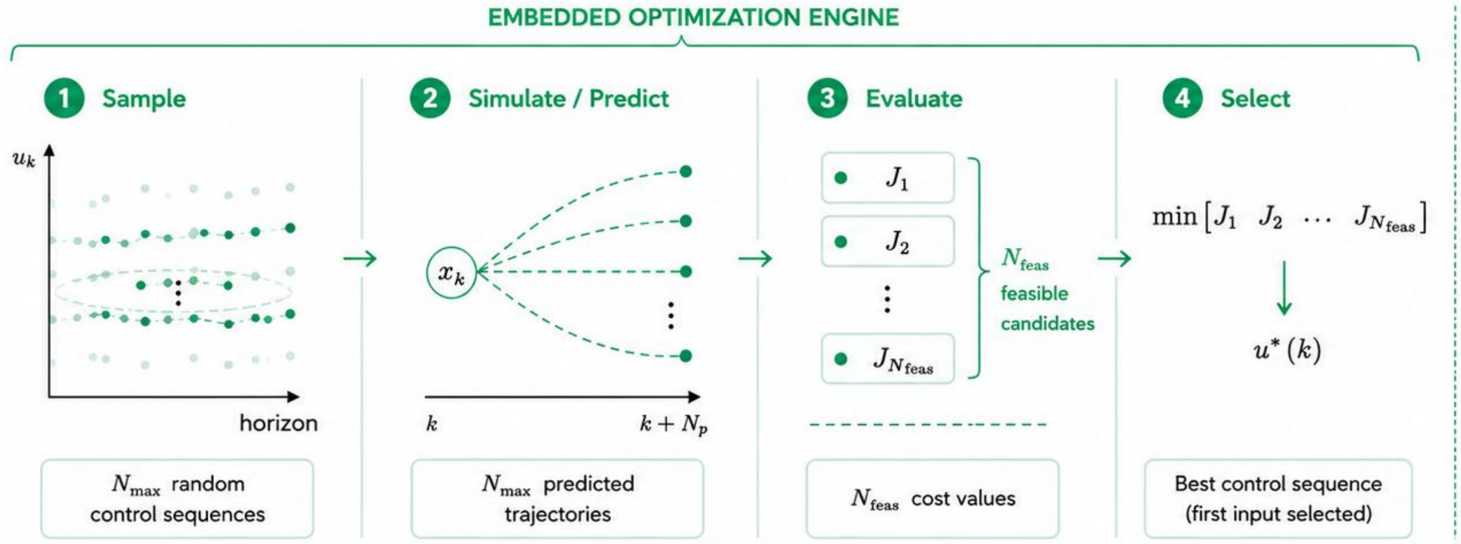

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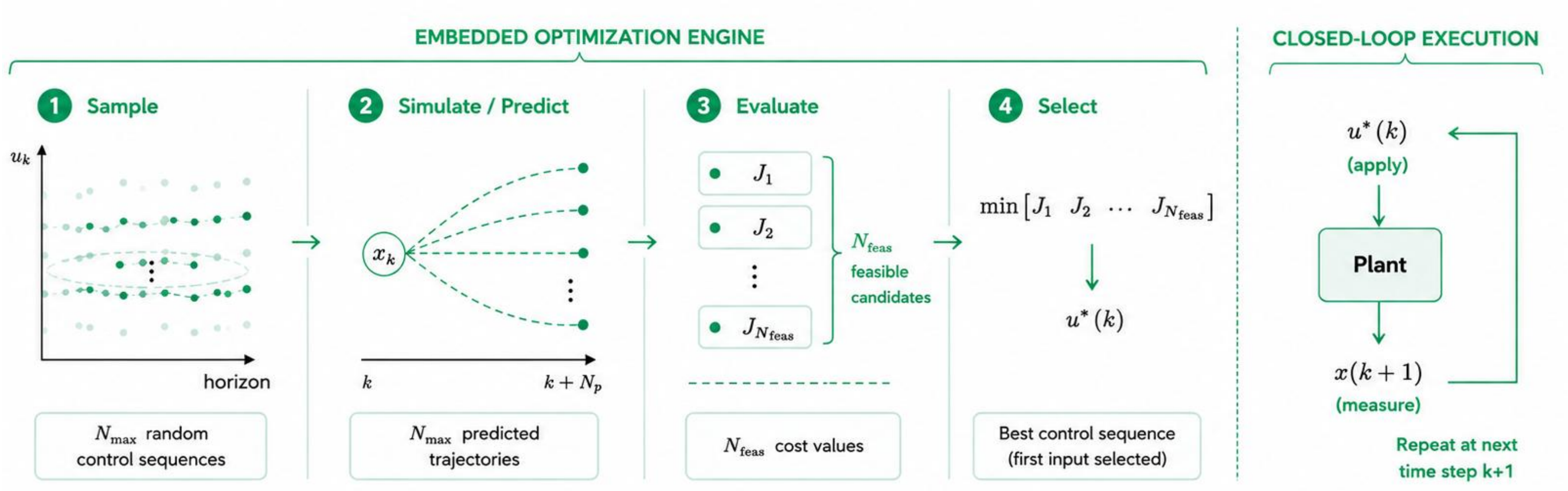


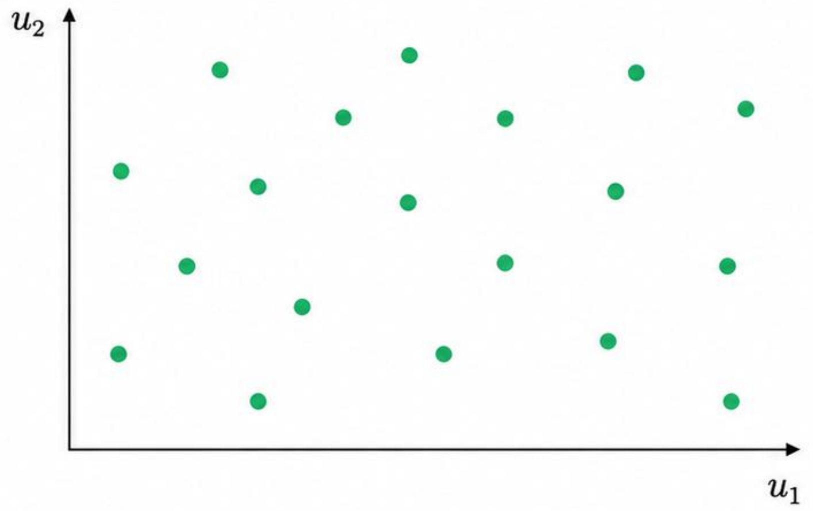
| | MPC |
|---|---|
| **Optimization** | |
| Optimization engine | QP solver |
| Runtime | Fixed |
| Memory | Moderate |

| | MPC |
|---|---|
| **Optimization** | |
| Optimization engine | QP solver |
| Runtime | Fixed |
| Memory | Moderate |
| **Control** | |
| Constraint handling | ✓ |
| Stability guarantees | ✓ |
| Closed-loop performance | Optimal |

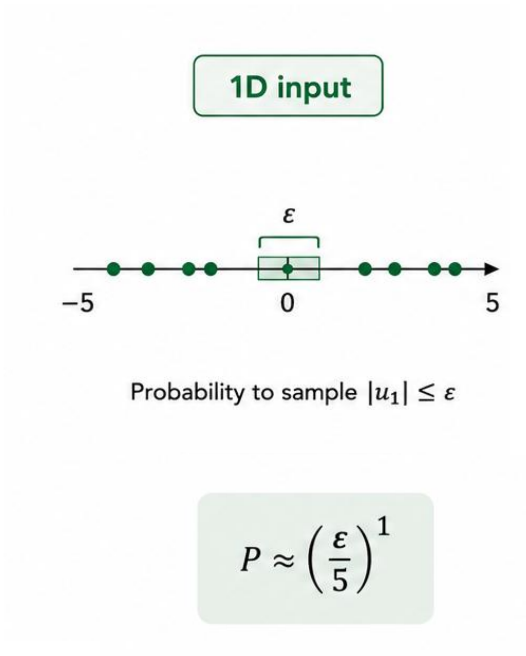
| | MPC | RS-MPC |
|---|---|---|
| **Optimization** | | |
| Optimization engine | QP solver | Random sampling |
| Runtime | Fixed | Adjustable |
| Memory | Moderate | Low |
| **Control** | | |
| Constraint handling | ✓ | ✓ |
| Stability guarantees | ✓ | ✗ |
| Closed-loop performance | Optimal | Baseline |

# Random Shooting

Samples are drawn uniformly across the input space.



# Sampling Near Optimum



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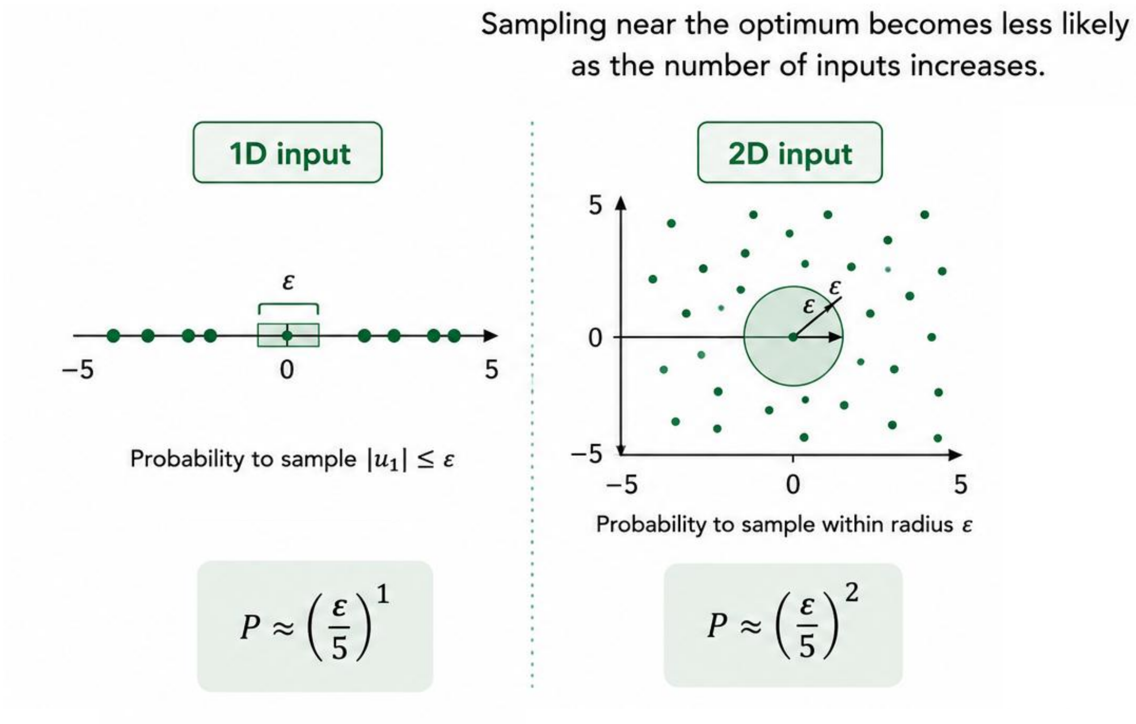

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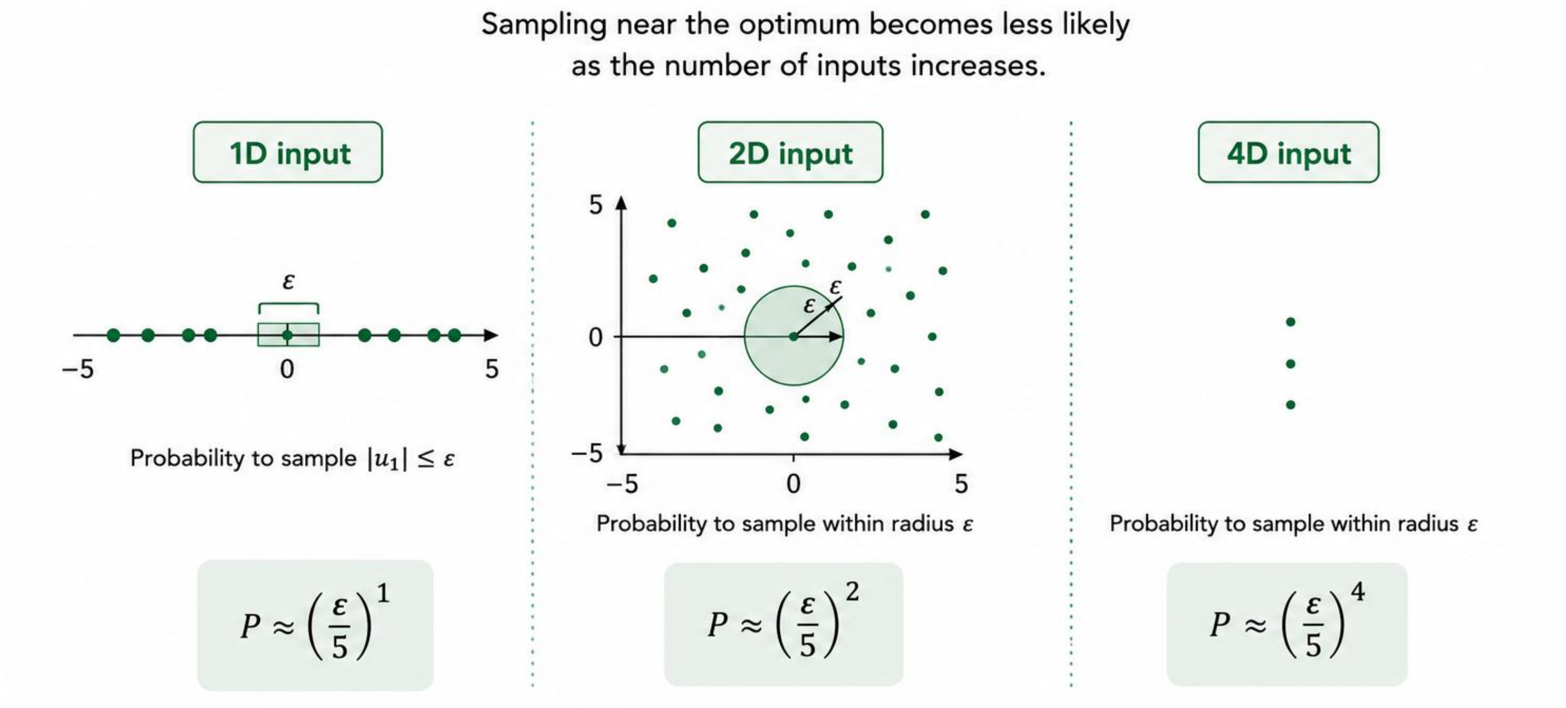

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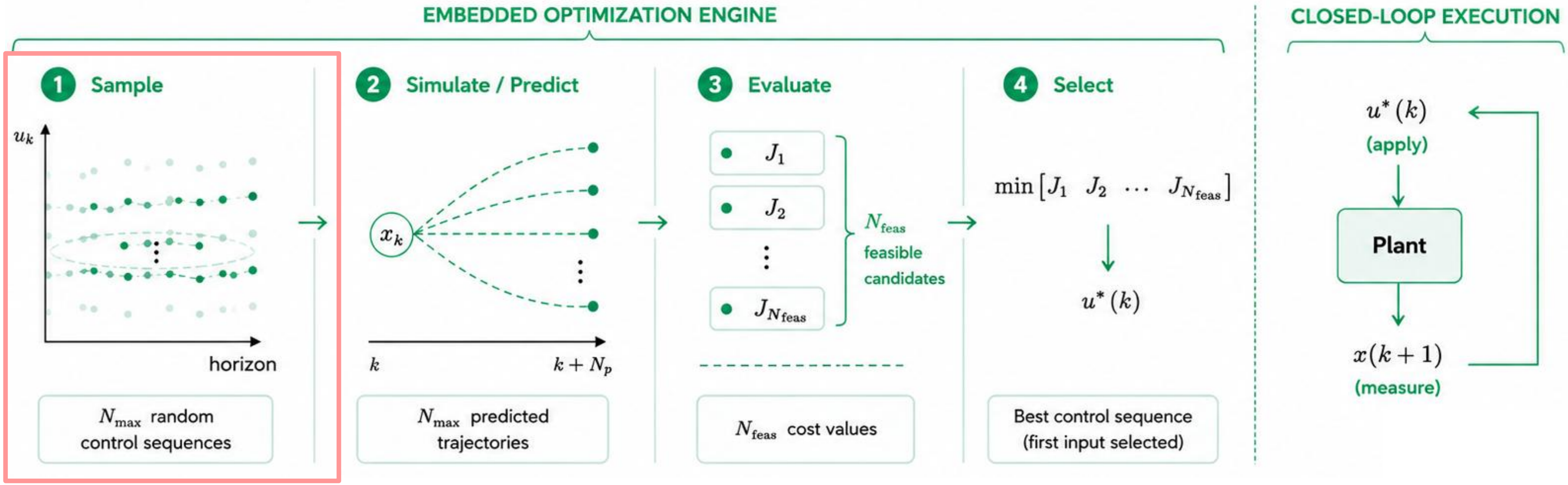

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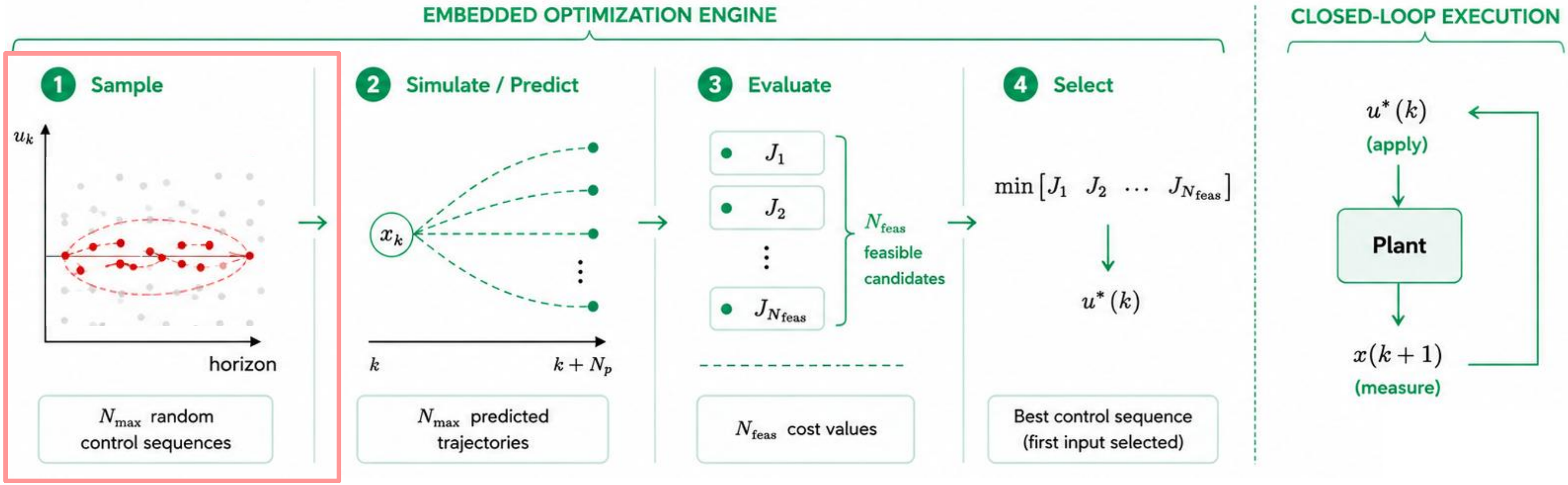
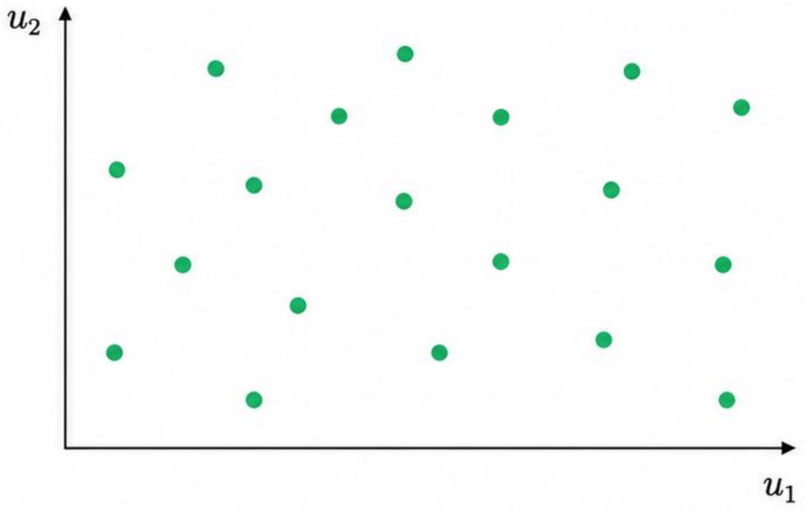


# Guided Sampling

## Random Shooting

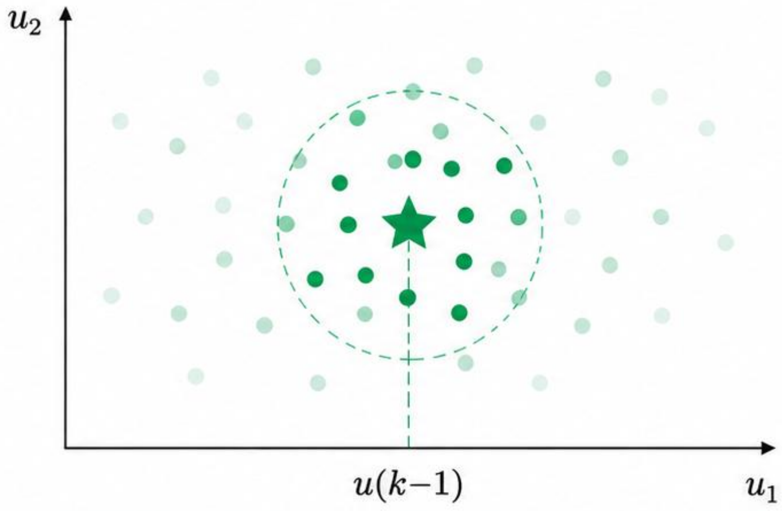
Samples are drawn uniformly across the input space.




Search everywhere

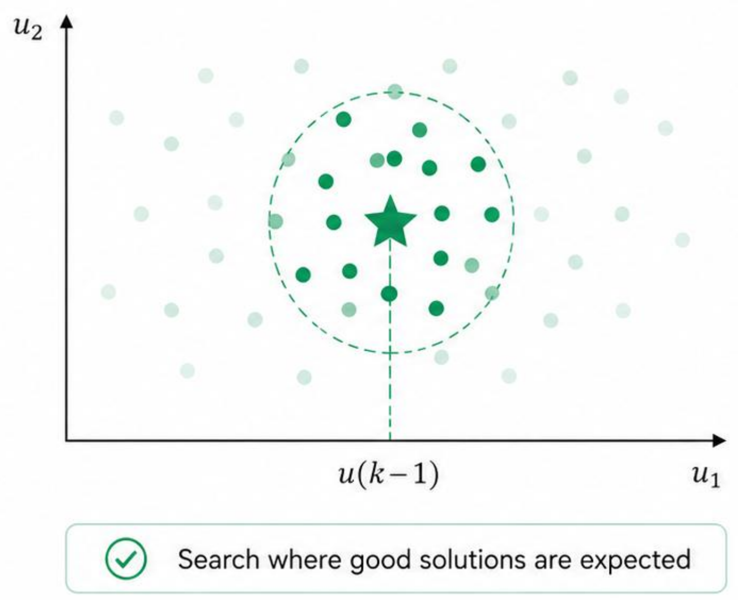
## Guided Sampling

Samples are drawn around the previous successful control action.



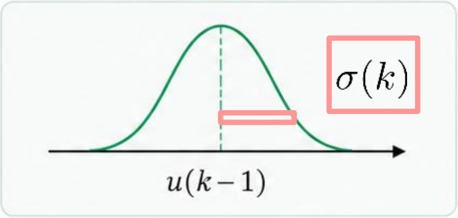

Search where good solutions are expected

# Guided Sampling: Gaussian Distribution

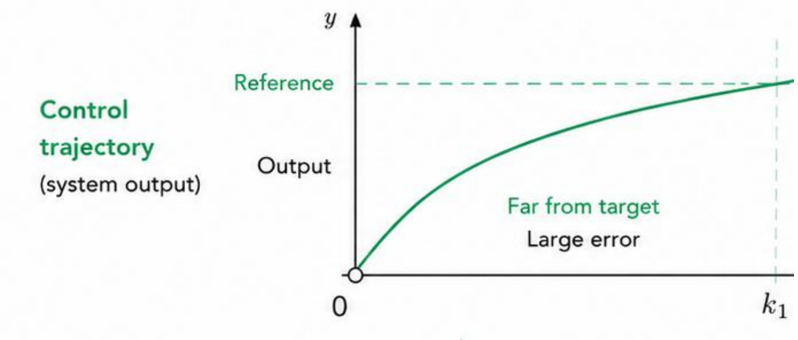



**Mean and Variance**

$$u(k) \sim \mathcal{N}(u(k-1),\, \sigma(k))$$



# Variance-Decaying Sampling



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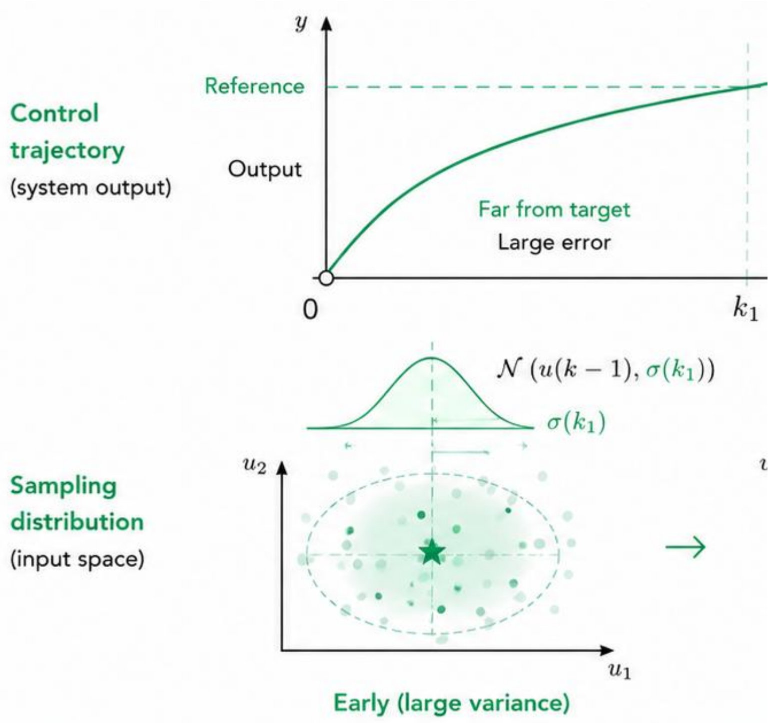

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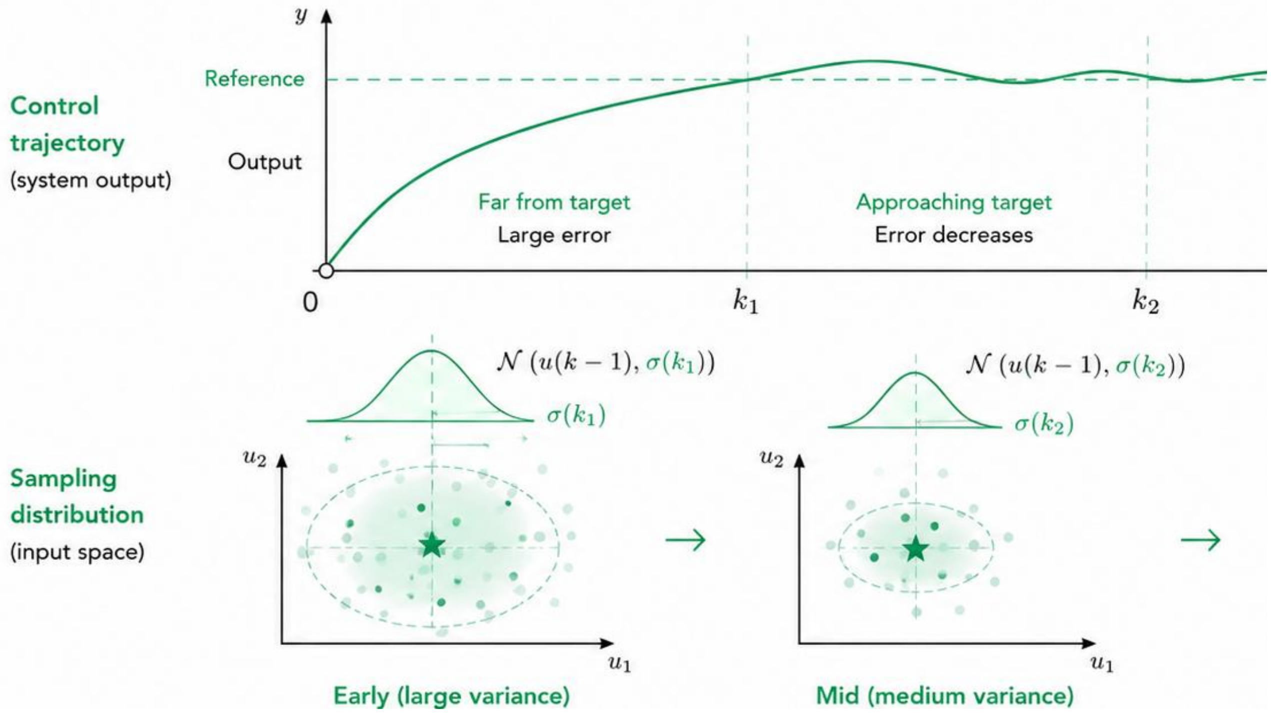

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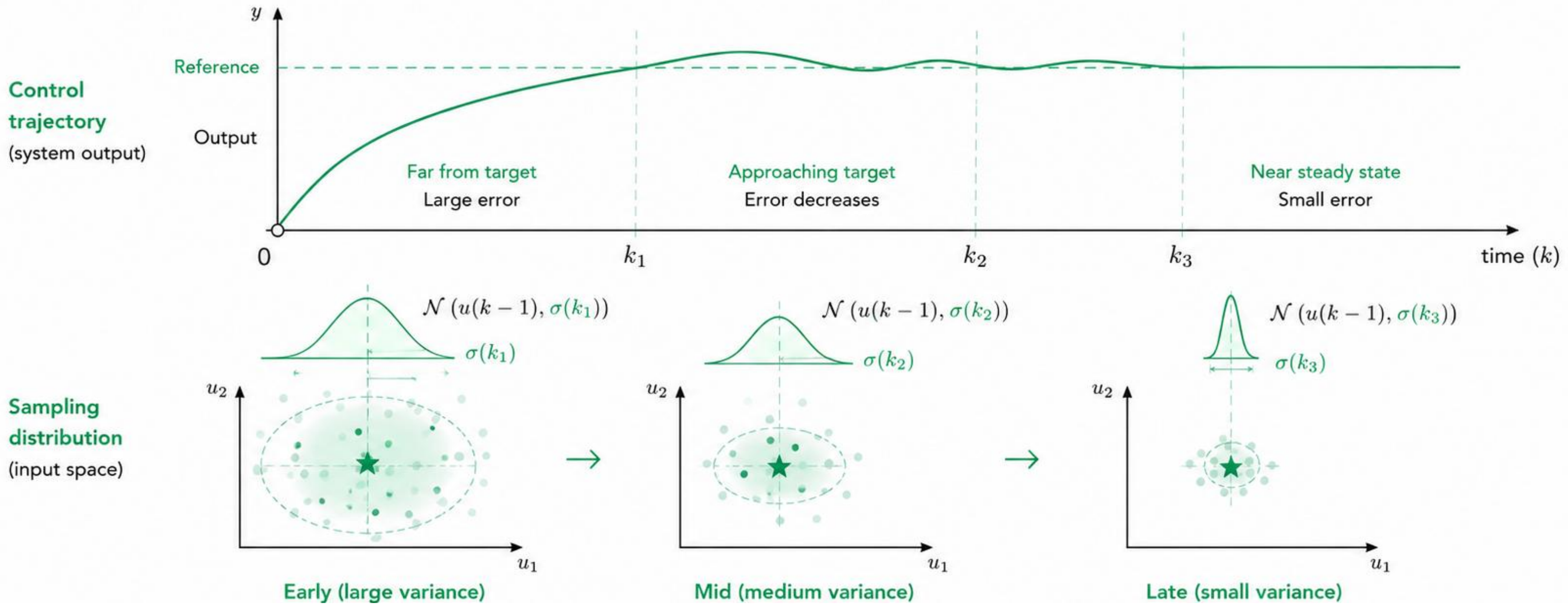
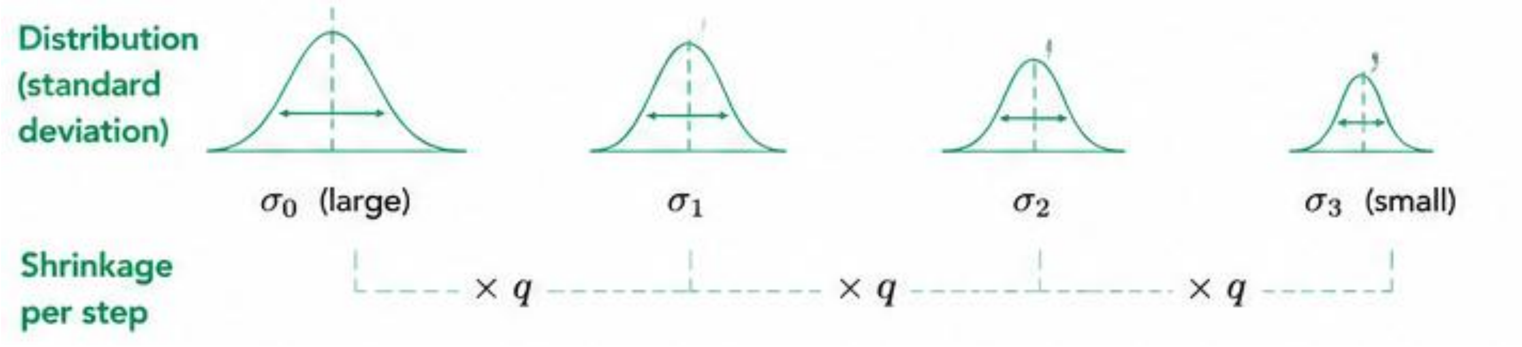
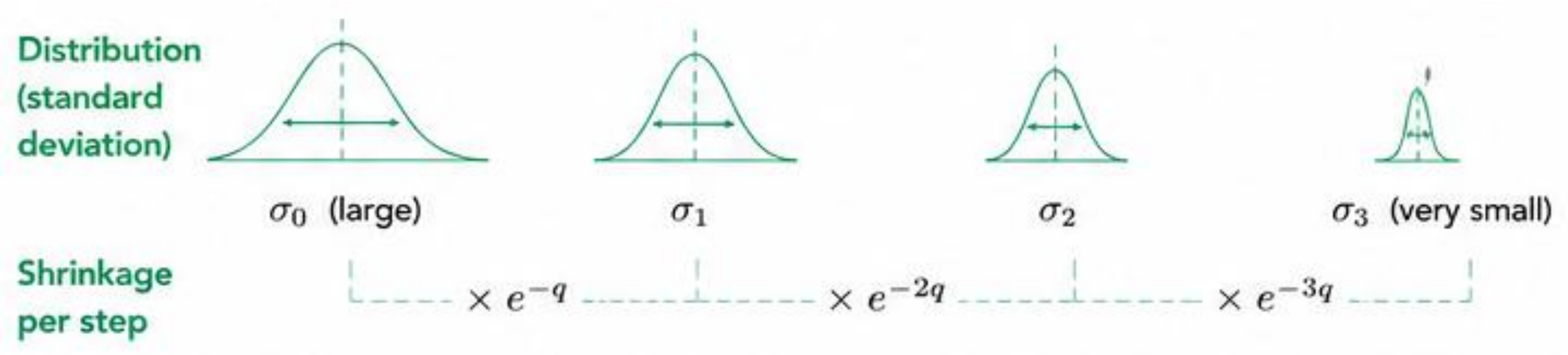


# Exponential Decay




$$\sigma(k) = q \cdot \sigma(k-1), \quad 0 < q < 1$$

# Geometric Decay




$$\sigma(k) = \sigma(k-1)e^{-q\,k}, \quad q > 0$$

| | MPC | RS-MPC | Proposed |
|---|---|---|---|
| **Optimization** | | | |
| Optimization engine | QP solver | Random sampling | **Guided sampling** |
| Runtime | Fixed | Adjustable | **Adjustable** |
| Memory | Moderate | Low | **Low** |
| **Control** | | | |
| Constraint handling | ✓ | ✓ | ✓ |
| Stability guarantees | ✓ | ✗ | ✓ |
| Closed-loop performance | Optimal | Baseline | **Improved** |

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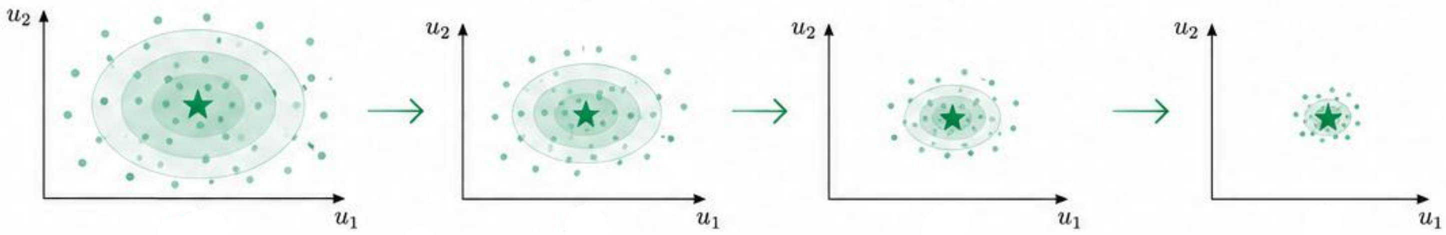
# Decaying Runtime

**Sampling distribution**

(input space)



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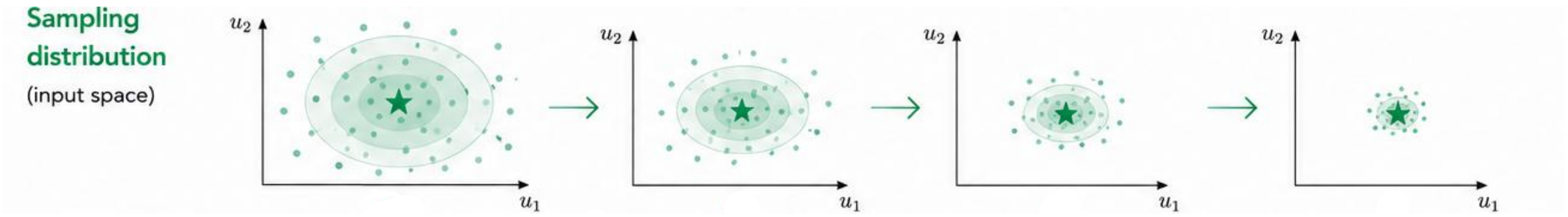


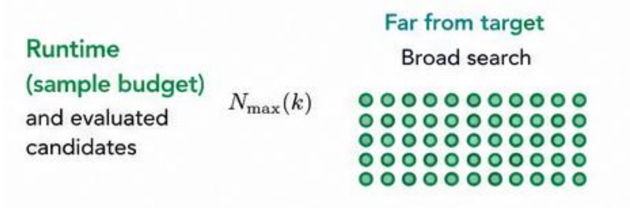

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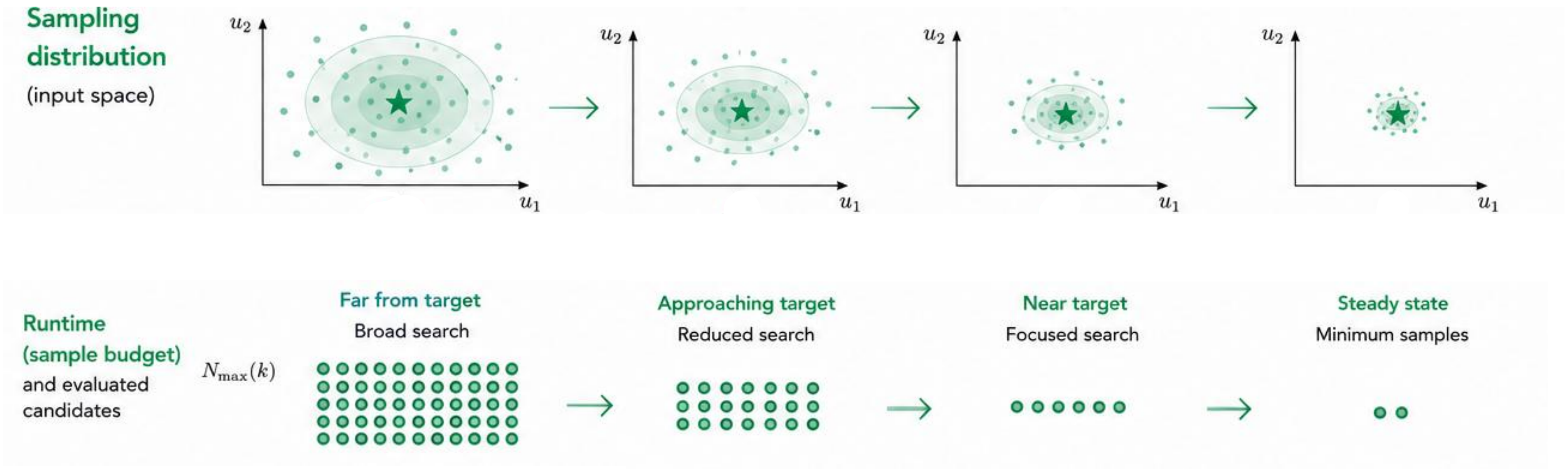

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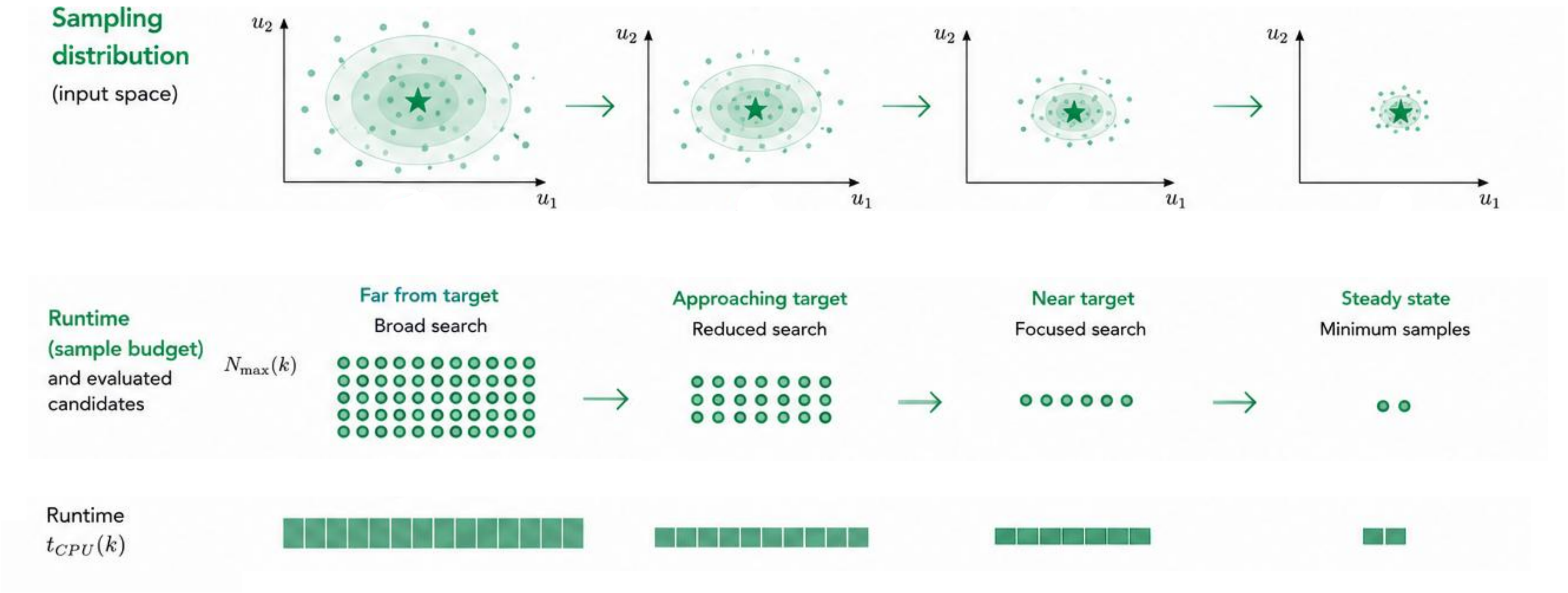
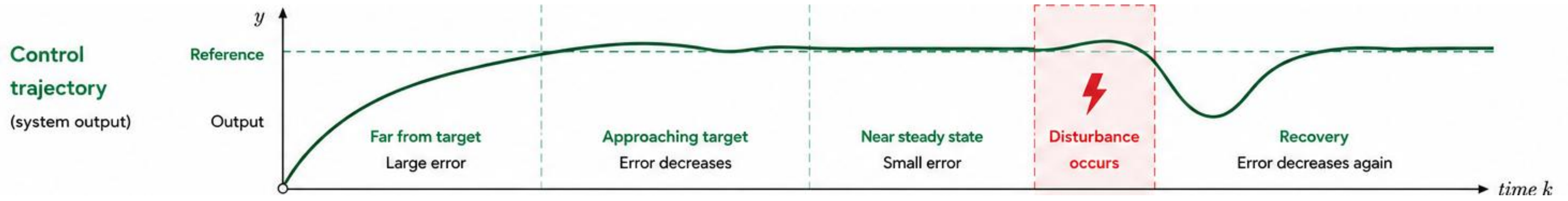


# Disturbance Handling



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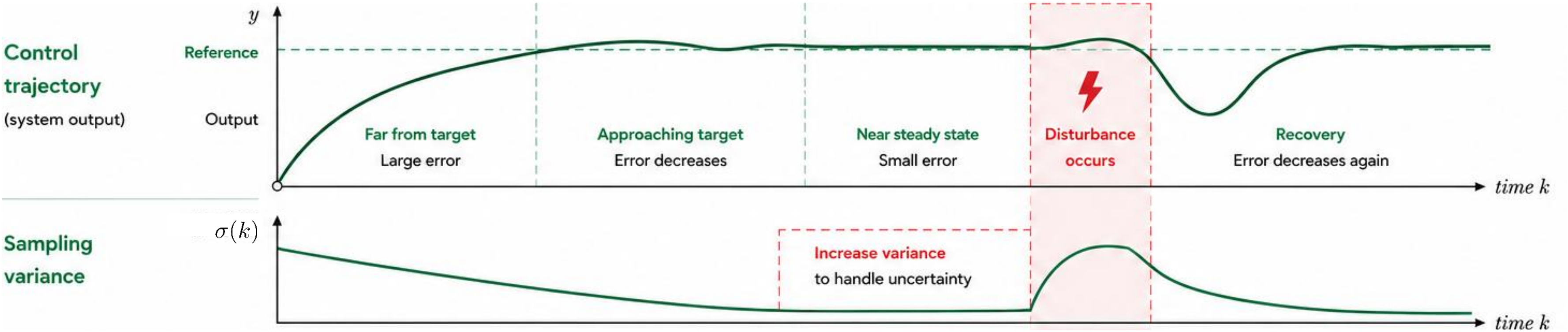

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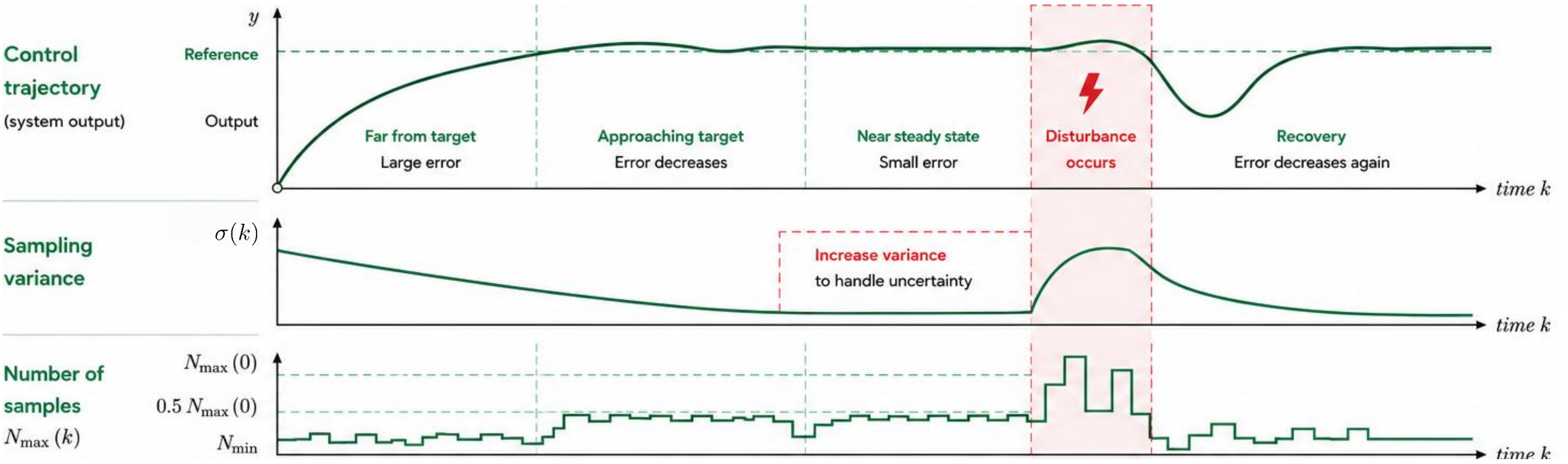

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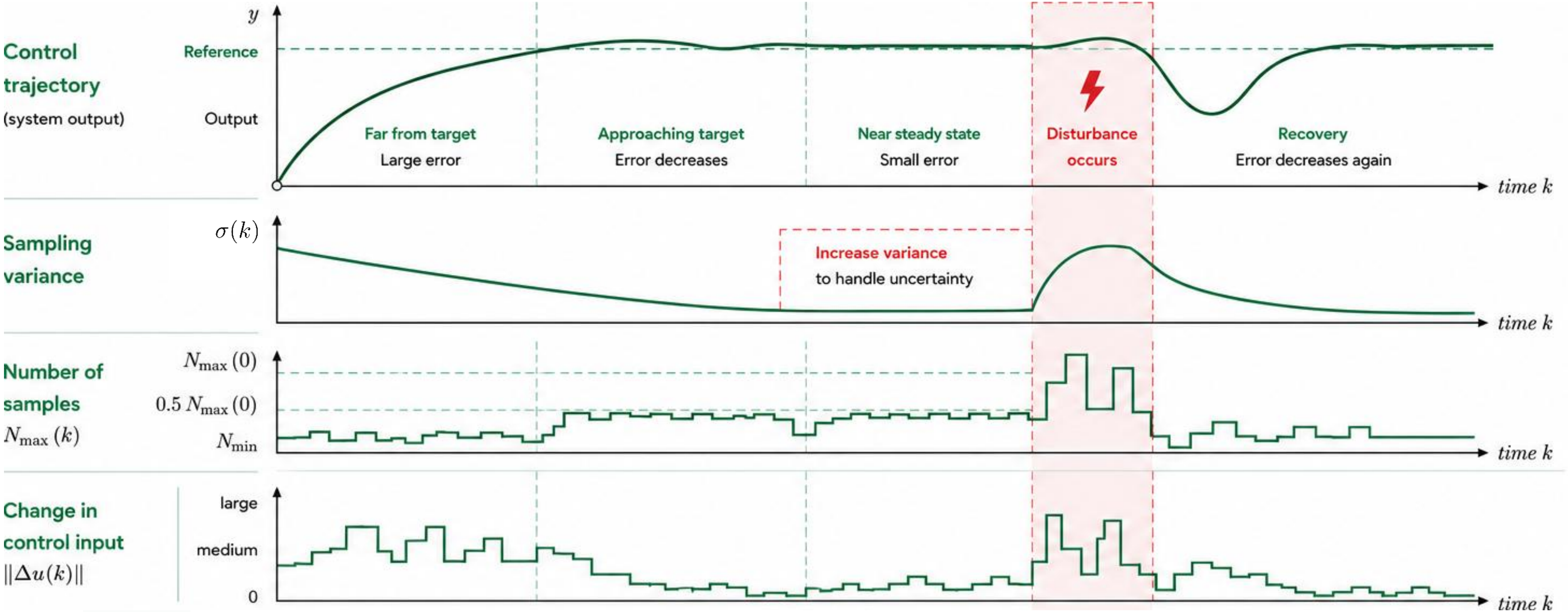
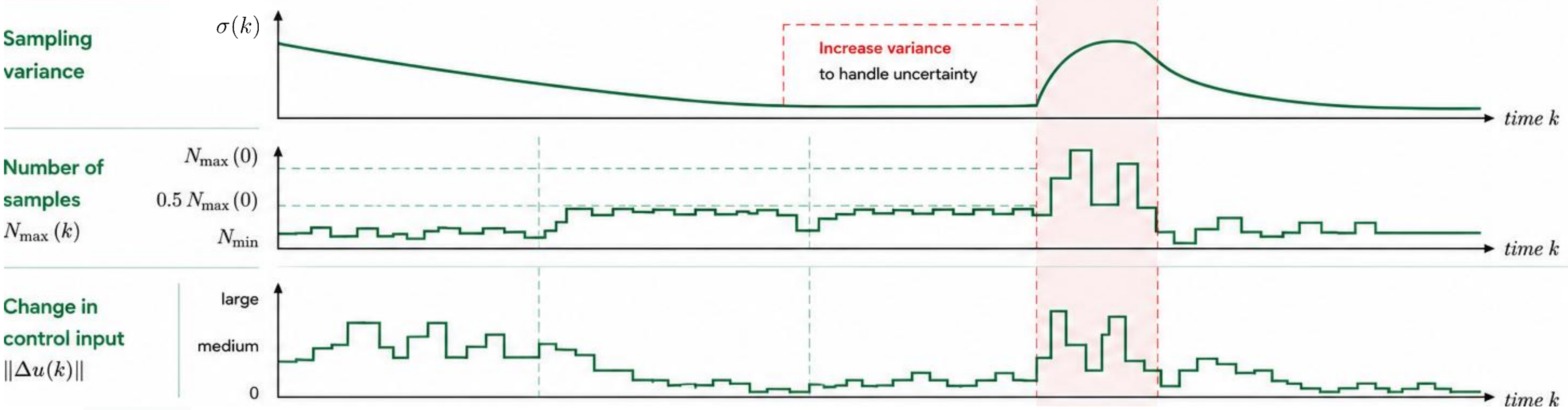
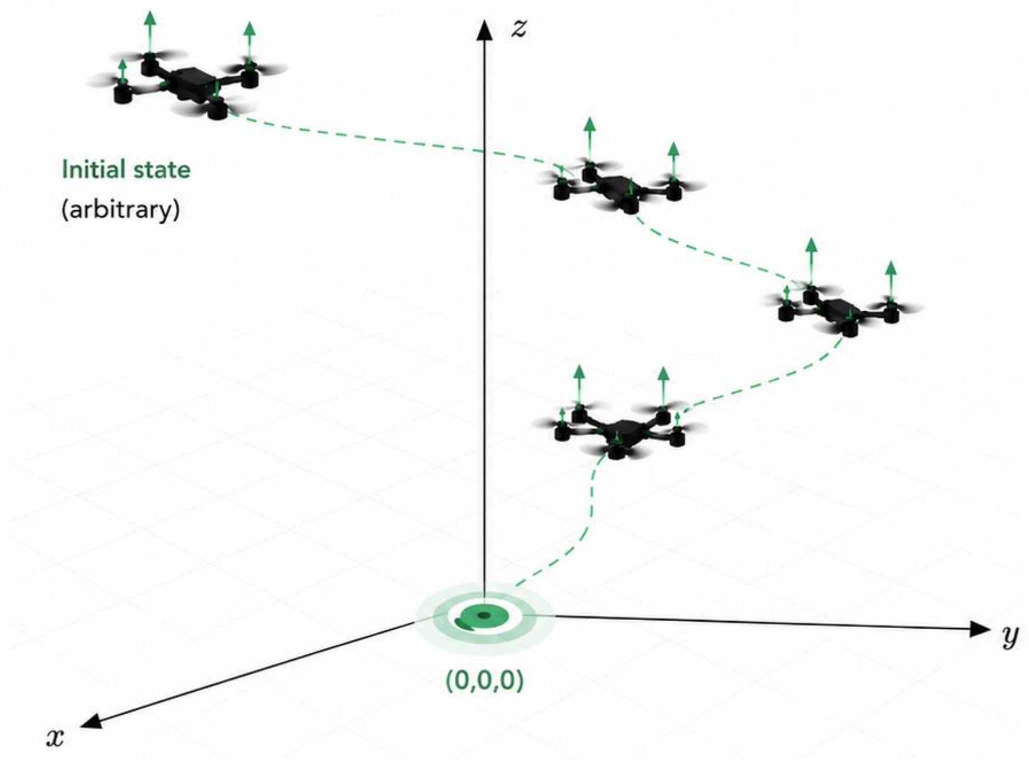


# Disturbance Handling

$$\underbrace{\Sigma(k)}_{\text{diagonal matrix}} = \text{diag}\left(\underbrace{q}_{\text{scaling factor}} \underbrace{|\Delta u(k)|}_{\text{magnitude of input change}}\right)$$



# Quadrotor Landing Benchmark




### Objective

Land at the origin (0,0,0)
with zero velocity.

$$x(k) \to 0, \;\; u(k) \to 0$$

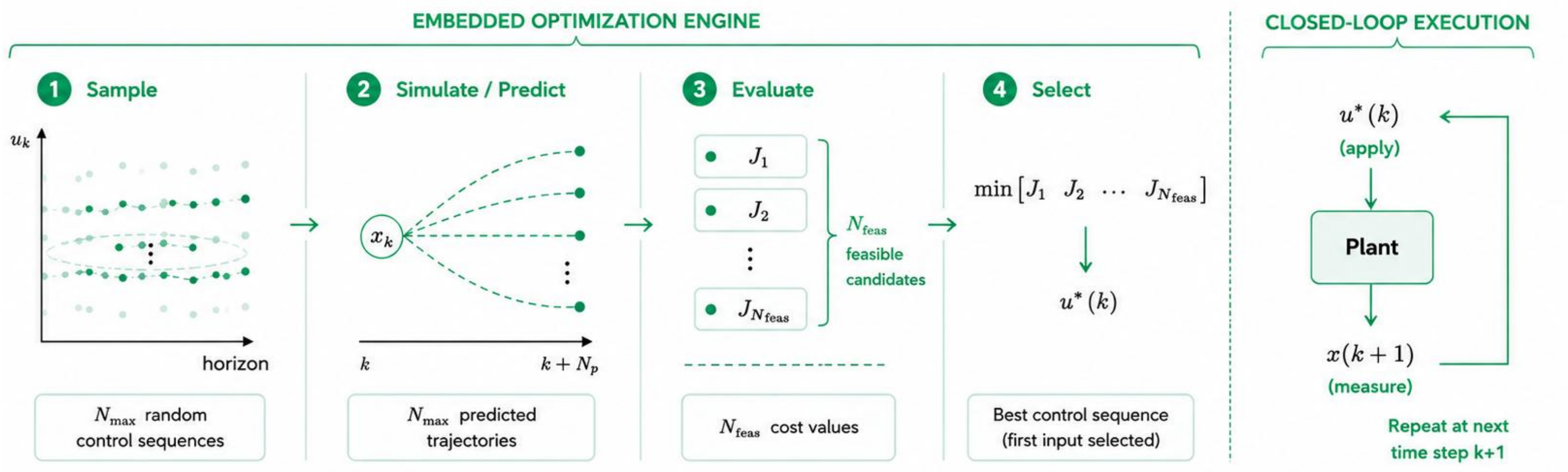
### Challenge

- Fast dynamics
- State and input constraints
- Limited onboard computational resources

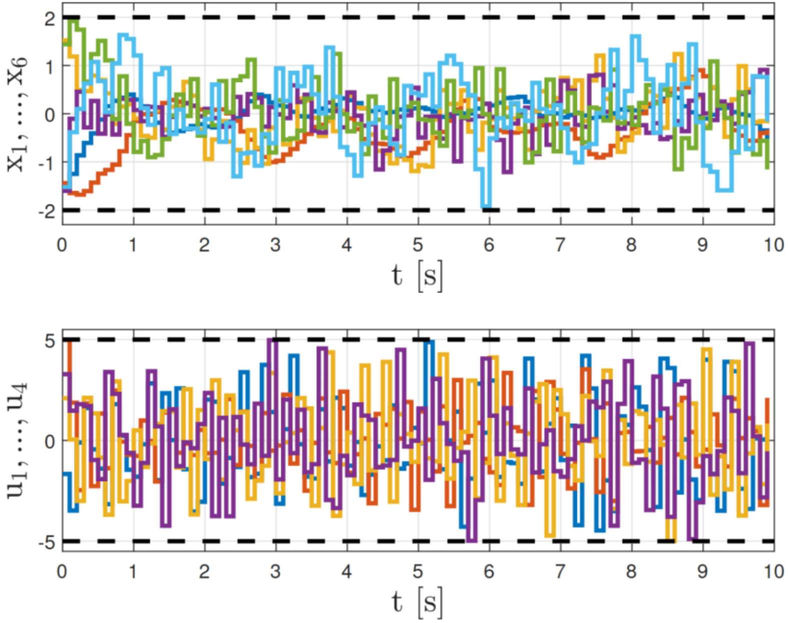
### Model

**6 states** (position, velocities, yaw) → **4 inputs** (velocity commands)

# Random-Shooting MPC



# Random-Shooting MPC: Quadrotor

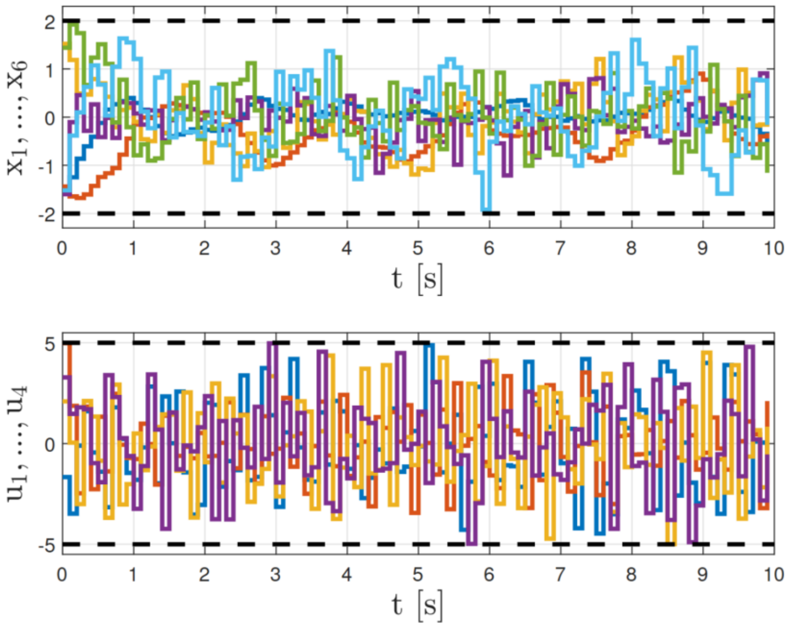



Probability to sample within radius $\epsilon$

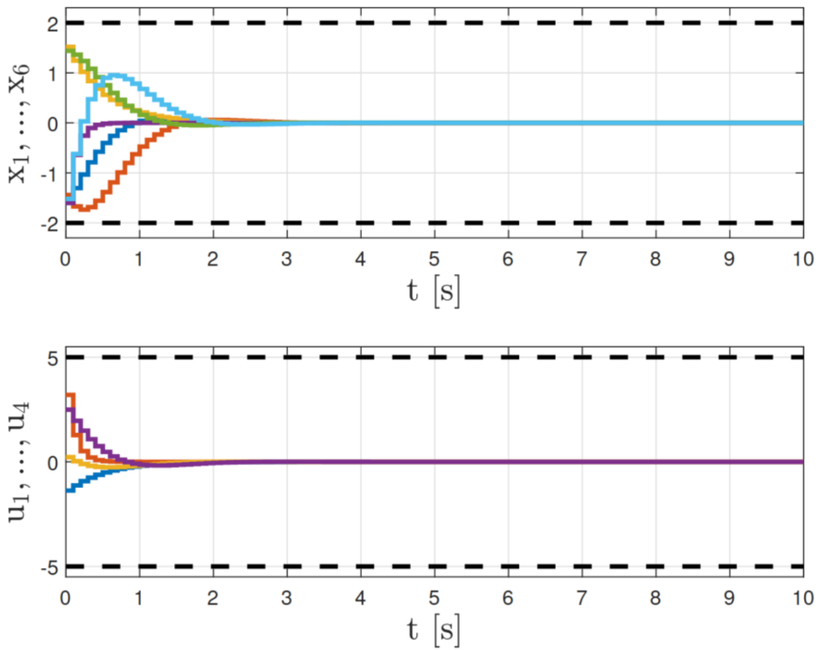
$$\approx \left(\frac{\epsilon}{5}\right)^4$$

(Exponentially smaller)

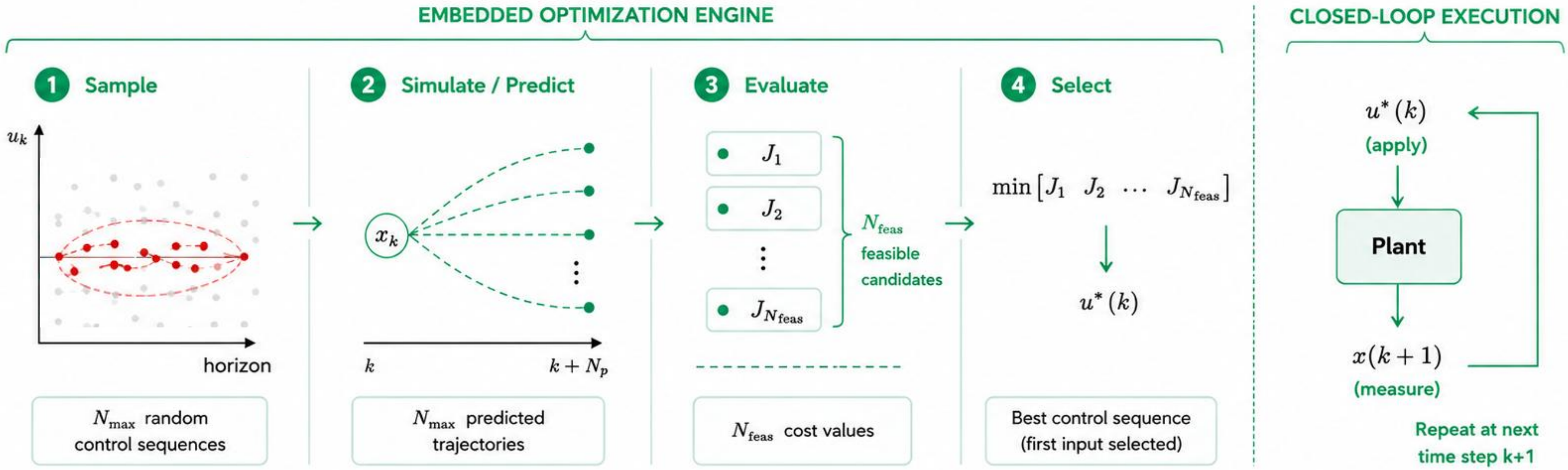
Random-Shooting MPC




MPC



# Variance-Decaying MPC



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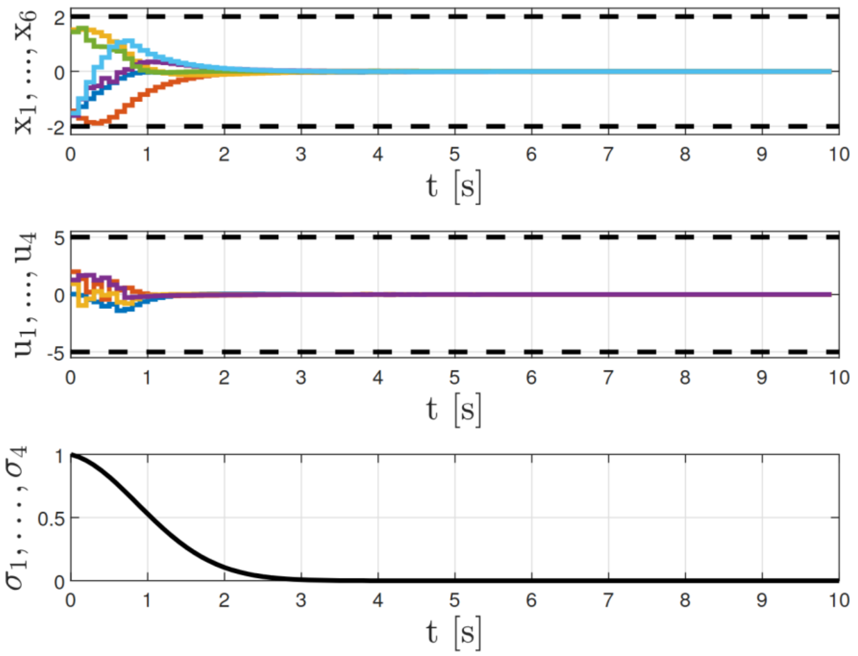
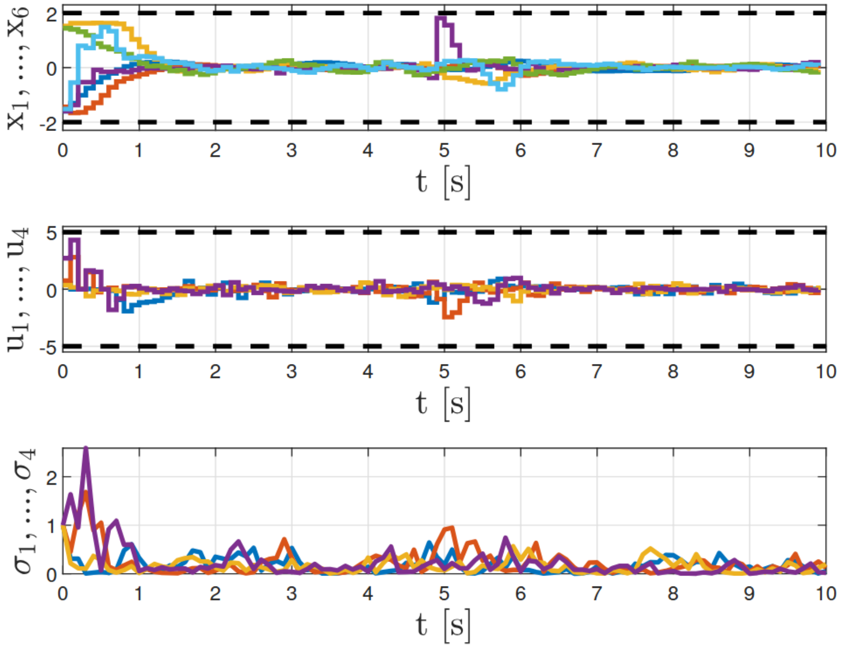


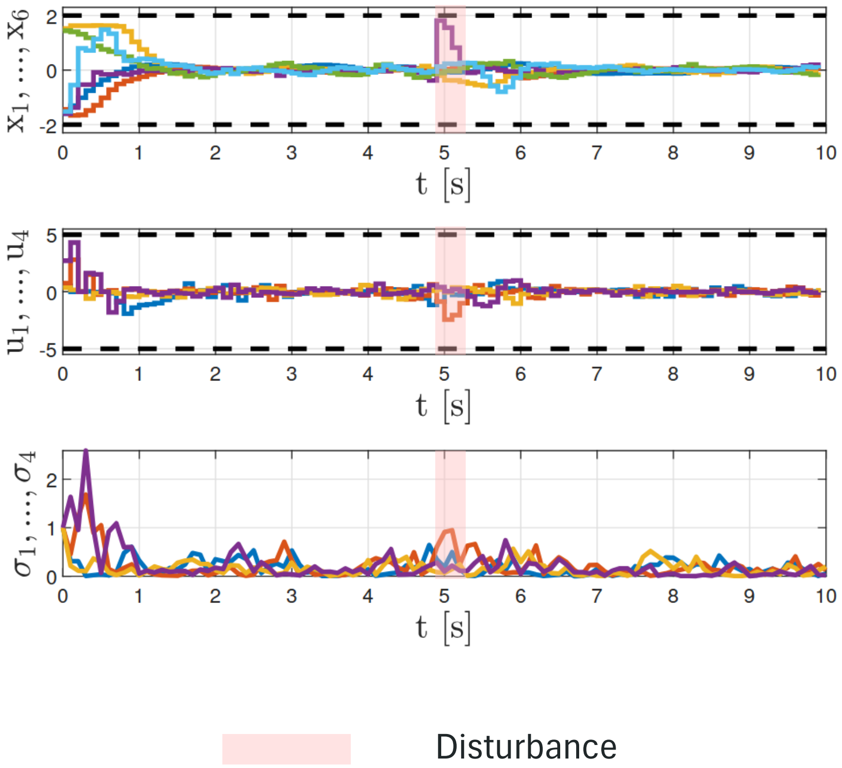

$$\sigma(k) = q \cdot \sigma(k-1), \quad 0 < q < 1,$$

# Variance-Adaptive MPC: Quadrotor

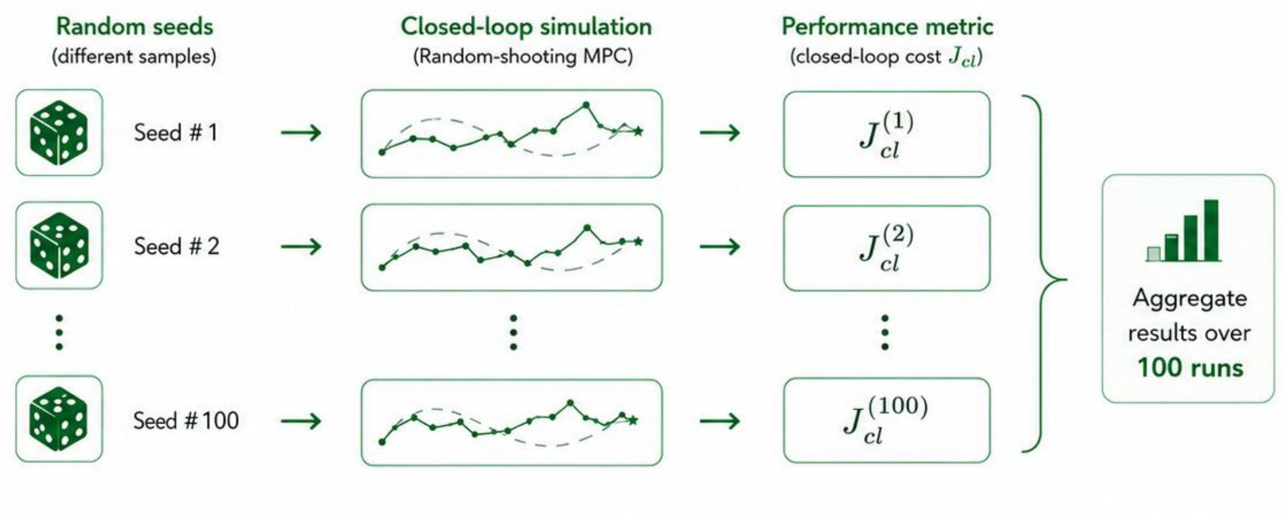



$$\sigma(k) = \mathrm{diag}\left(q \left|\Delta u(k)\right|\right)$$

# Variance-Adaptive MPC: Quadrotor



# 100 Monte Carlo Runs



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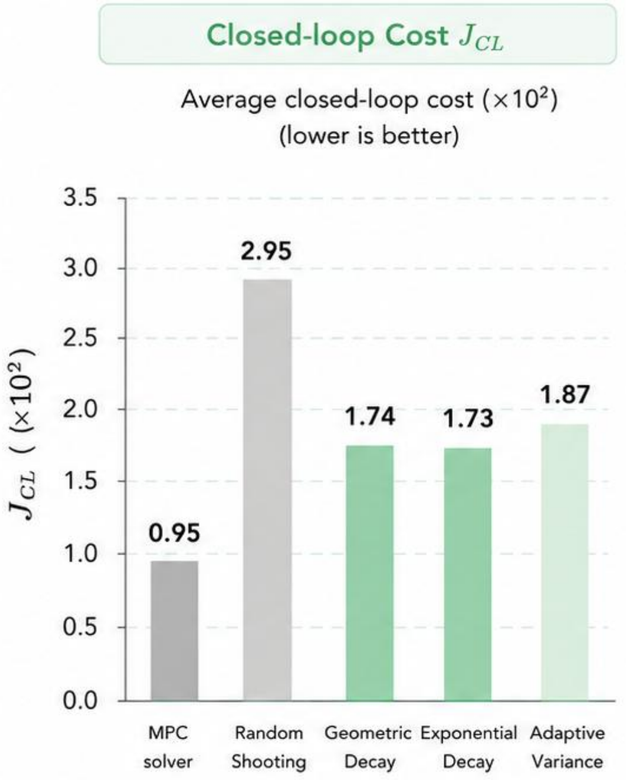

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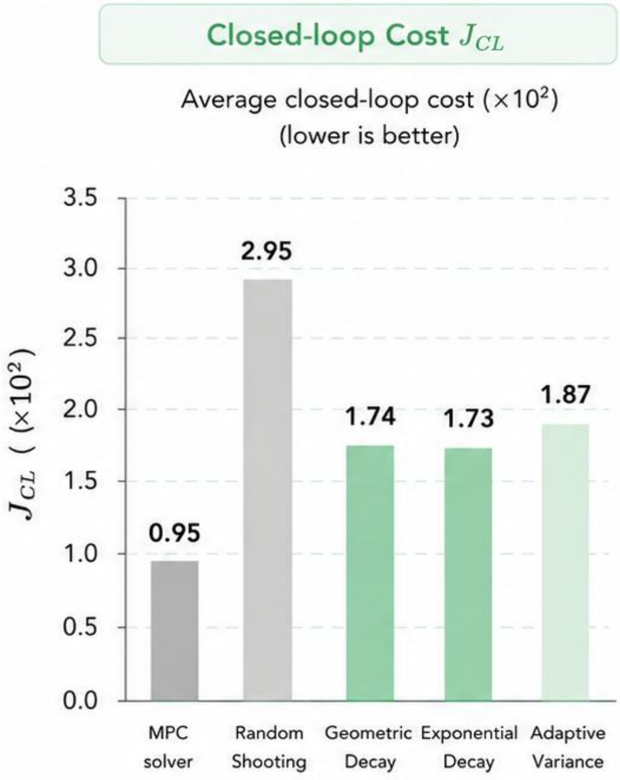
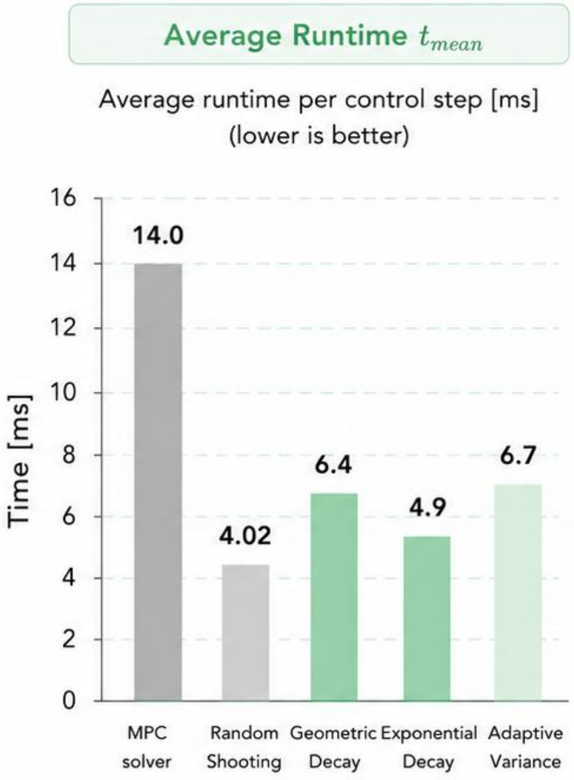
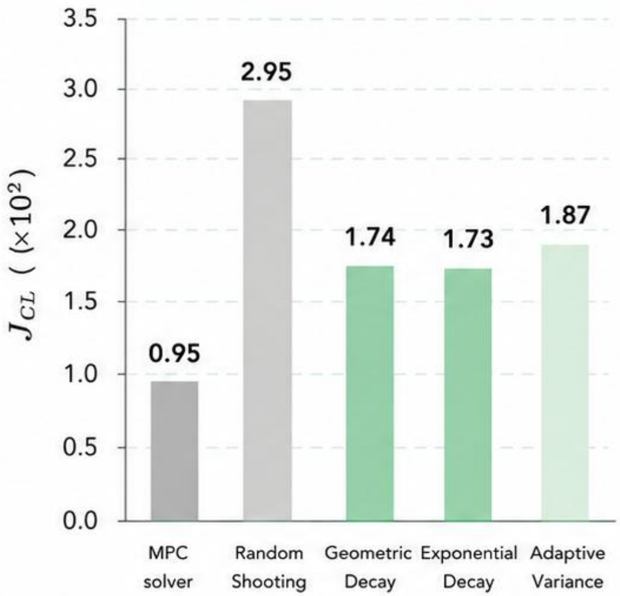


# 100 Monte Carlo Runs

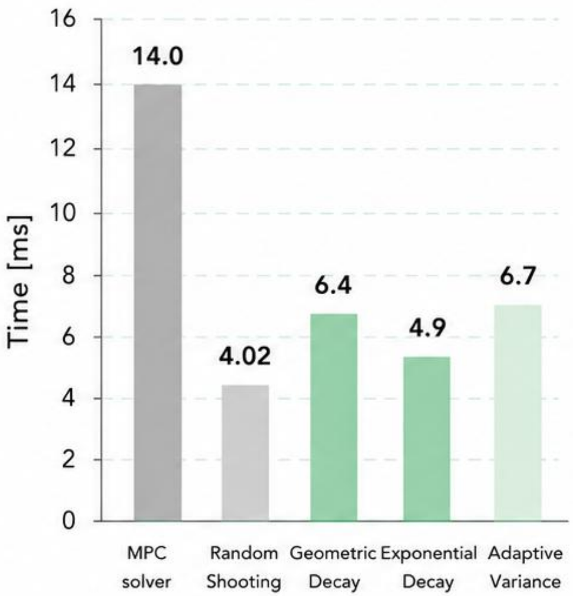
**Closed-loop Cost $J_{CL}$**

Average closed-loop cost ($\times 10^2$)
(lower is better)



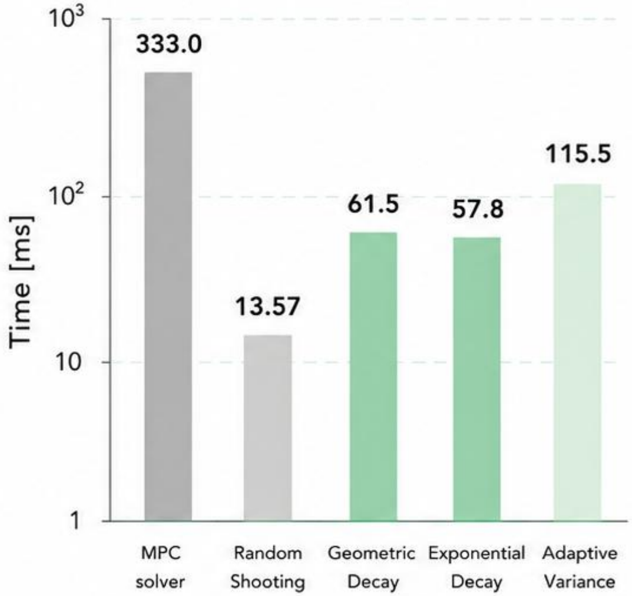

**Average Runtime $t_{mean}$**

Average runtime per control step [ms]
(lower is better)

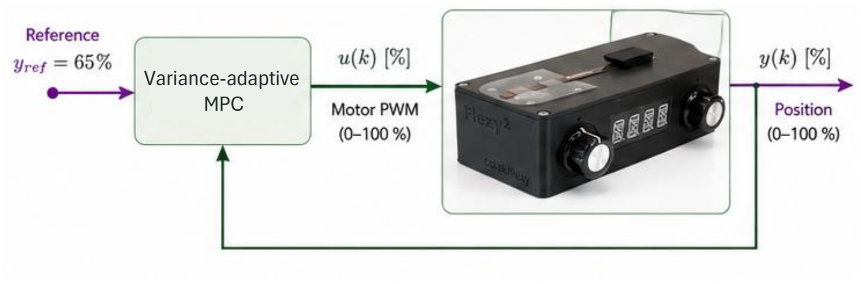


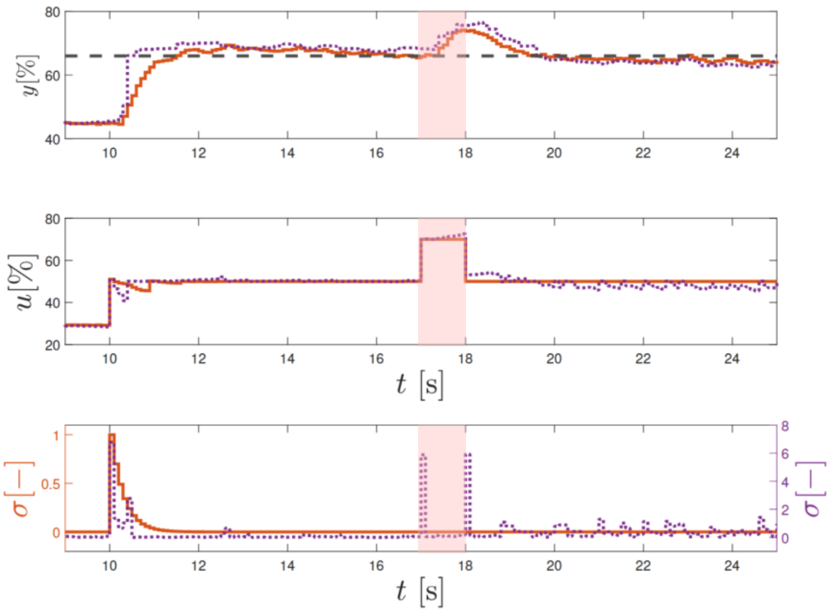

**Maximal Runtime $t_{max}$**

Maximal runtime per control step [ms]
(lower is better)



# Experimental Validation




Disturbance

# Conclusions

| | MPC | RS-MPC | Proposed |
|---|---|---|---|
| **Optimization** | | | |
| Optimization engine | QP solver | Random sampling | **Guided sampling** |
| Average runtime (ms) | 14.0 | 4.02 | **6.4** (−54.3% vs MPC) |
| Closed-loop cost $J_{CL}$ ($\times 10^2$) | 0.95 | 2.95 | **1.87** (−36.6% vs RS-MPC) |
| **Control** | | | |
| Constraint handling | ✓ | ✓ | ✓ |
| Stability guarantees | ✓ | ✗ | ✓ |
| Overall performance | Optimal | Baseline | **Improved** (better cost, similar runtime) |

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# Acknowledgement

M. Horváthová acknowledges the contribution of the EU NextGenerationEU through the Recovery and Resilience Plan for Slovakia under project No. 09I03-03-V04-00636